Vector Operations

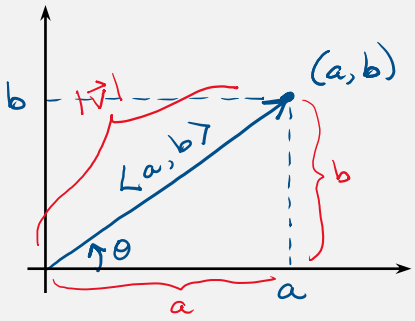
Vector Components, Direction, and Magnitude

So far, we have introduced vectors from a geometric perspective. Now, we'll bring them into the coordinate plane to extend our understanding and perform operations with vectors. In this lesson, we break down vectors into component form, discuss how to calculate magnitude and direction, and introduce rules for operations involving vectors.

DEFINITION: ***Position Vectors, Vector Components, Direction Angle and Magnitude***

A vector $\mathbf{v}$ with initial point (tail) placed at the origin and terminal point (head) at the coordinate (a, b) is called a position vector, and is denoted as $\langle a, b \rangle$.

The real numbers a and b are the horizontal and vertical components, respectively, of the vector $\mathbf{v}$.



The positive angle formed with initial side along the positive x- axis and terminal side formed by the vector $\mathbf{v}$ is called the direction angle. This angle satisfies the equation $\tan\theta = \frac{b}{a}, \; a \neq 0$

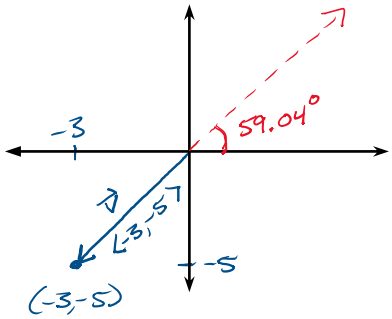
The magnitude or length of the vector $\mathbf{v} = \langle a, b \rangle$ is denoted as $|\vec{v}|$ and can be calculated by the equation, $|\vec{v}| = \sqrt{a^2 + b^2}$.

EXAMPLE 1

Let $\mathbf{v} = \langle -3, -5 \rangle$. Graph $\mathbf{v}$, evaluate $|\mathbf{v}|$, and find the direction angle.

$$|\vec{v}| = |\langle -3, -5 \rangle|$$
$$= \sqrt{(-3)^2 + (-5)^2}$$
$$= \sqrt{34}$$

$$\tan\theta = \frac{-5}{-3} = \frac{5}{3}$$
$$\theta' = \tan^{-1}\left(\frac{5}{3}\right) \approx 59.04^\circ$$
$$\theta = \tan^{-1}\left(\frac{5}{3}\right) + 180^\circ \approx 239.04^\circ$$



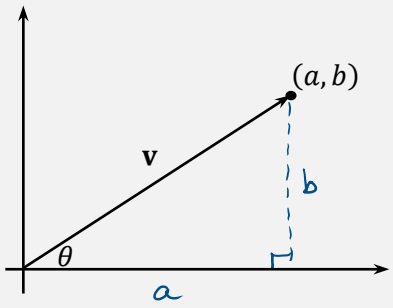
IMPORTANT INFO: *Finding Vector Components*

A vector $\mathbf{v}$, with magnitude $|\mathbf{v}|$ and direction angle θ, has horizontal and vertical components given by the following:

$\sin\theta = \frac{b}{|\vec{v}|} \rightarrow b = |\vec{v}|\sin\theta$ (vert. comp.)

$\cos\theta = \frac{a}{|\vec{v}|} \rightarrow a = |\vec{v}|\cos\theta$ (horiz. comp.)

$\vec{v} = \langle a, b\rangle = \langle |\vec{v}|\cos\theta, |\vec{v}|\sin\theta\rangle$



EXAMPLE 2

For each description below, determine the components of the given vector and express as a position vector.

(a) Vector $\mathbf{u}$ with magnitude 7 and direction angle of 60°.

$\theta = 60°$ $\quad \vec{u} = \langle 7\cos(60°), 7\sin(60°)\rangle$

$|\vec{u}| = 7$ $\quad \vec{u} = \langle \frac{7}{2}, \frac{7\sqrt{3}}{2}\rangle$

(b) Vector $\mathbf{v}$ with magnitude 5 and direction angle of 222°.

$\theta = 222°$ $\quad \vec{v} = \langle 5\cos(222°), 5\sin(222°)\rangle$

$|\vec{v}| = 5$ $\quad \vec{v} \approx \langle -3.72, -3.35\rangle$

(c) Vector $\mathbf{w}$ with magnitude 2π and direction angle of 180°.

$\theta = 180°$ $\quad \vec{w} = \langle 2\pi\cos(180°), 2\pi\sin(180°)\rangle$

$|\vec{w}| = 2\pi$ $\quad \vec{w} = \langle -2\pi, 0\rangle$

Vector Operations

DEFINITION: *Vector Addition, Subtraction, and Scalar Multiplication*

Let $\mathbf{v} = \langle a, b \rangle$ and $\mathbf{w} = \langle c, d \rangle$; we define ***vector addition*** and ***subtraction*** as follows.

$$\mathbf{v} \pm \mathbf{w} = \langle a, b \rangle \pm \langle c, d \rangle = \langle a \pm c, b \pm d \rangle$$

Let k be a real value; we define ***scalar multiplication*** as follows.

$$k\mathbf{v} = k\langle a, b \rangle = \langle ka, kb \rangle$$

EXAMPLE 3

Let $\mathbf{v} = \langle 1, -2 \rangle$ and $\mathbf{w} = \langle -2, 3 \rangle$. For each expression, state whether the result will be a scalar or vector quantity and evaluate.

$-2\mathbf{v}$

$-2\langle 1, -2 \rangle$

$= \langle -2(1), -2(-2) \rangle$

$= \langle -2, 4 \rangle$, vector

(opposite direction of $\vec{v}$ with $\times 2$ magnitude)

$-2\mathbf{v} - \mathbf{w}$

$\langle -2, 4 \rangle - \langle -2, 3 \rangle$

$= \langle -2 - -2, 4 - 3 \rangle$

$= \langle 0, 1 \rangle$, vector

$|-2\mathbf{v} - \mathbf{w}|$

$|\langle 0, 1 \rangle| = \sqrt{0^2 + 1^2}$

$= 1$, scalar

$|\langle 0, 1 \rangle| = 1$ so we call this a unit vector

EXAMPLE 4

Express the vector $\langle \frac{3}{5}, \frac{4}{5} \rangle$ as a sum of vectors $\mathbf{v} = \langle 1, 0 \rangle$ and $\mathbf{w} = \langle 0, 1 \rangle$, then evaluate $\left| \langle \frac{3}{5}, \frac{4}{5} \rangle \right|$.

(unit vector) (unit vector)

$\langle \frac{3}{5}, \frac{4}{5} \rangle = \langle \frac{3}{5}, 0 \rangle + \langle 0, \frac{4}{5} \rangle$

$= \frac{3}{5}\langle 1, 0 \rangle + \frac{4}{5}\langle 0, 1 \rangle$

$= \frac{3}{5}\vec{v} + \frac{4}{5}\vec{w}$

(a linear combination of unit vectors)

$\left| \langle \frac{3}{5}, \frac{4}{5} \rangle \right| = \sqrt{(3/5)^2 + (4/5)^2}$

$= \sqrt{9/25 + 16/25}$

$= \sqrt{25/25}$

$= 1$

Standard Basis Vectors

DEFINITION: *Unit Vector and Standard Basis Vectors*

A unit vector is any vector with magnitude 1. Unit vectors are often used to indicate direction.

There are two convenient unit vectors, $\hat{\imath} = \langle 1,0 \rangle$ and $\hat{\jmath} = \langle 0,1 \rangle$, read aloud as i-hat and j-hat, and are sometimes referred to as the standard basis vectors.

These unit vectors are perpendicular (or orthogonal) and form a ***standard basis*** for the xy-plane. That is, any vector in the plane can be expressed as a combination of these two perpendicular unit vectors, $\hat{\imath}$ and $\hat{\jmath}$.

EXAMPLE 5

Let $\mathbf{v} = \langle 1, -2 \rangle$ and $\mathbf{w} = \langle -2, 3 \rangle$.

(a) Express each vector $\mathbf{v}$ and $\mathbf{w}$ as a combination of the standard basis vectors, $\hat{\imath}$ and $\hat{\jmath}$.

$\vec{v} = \langle 1, -2 \rangle = 1\langle 1,0 \rangle - 2\langle 0,1 \rangle = 1\hat{\imath} - 2\hat{\jmath}$

$\vec{w} = \langle -2, 3 \rangle = -2\hat{\imath} + 3\hat{\jmath}$

(b) Express each quantity below as a combination of standard basis vectors, $\hat{\imath}$ and $\hat{\jmath}$.

$-2\mathbf{v}$

$-2(\hat{\imath} - 2\hat{\jmath}) = -2\hat{\imath} + 4\hat{\jmath}$

$-2\mathbf{v} - \mathbf{w}$

$-2\hat{\imath} + 4\hat{\jmath} - (-2\hat{\imath} + 3\hat{\jmath})$

$= -2\hat{\imath} + 4\hat{\jmath} + 2\hat{\imath} - 3\hat{\jmath}$

$= -2\hat{\imath} + 2\hat{\imath} + 4\hat{\jmath} - 3\hat{\jmath}$

$= \hat{\jmath}$

The Dot Product

When working with scalar quantities, we have just one form of multiplication. However, with vector quantities, there are several forms of multiplication. We've introduced one simple version already, scalar multiplication. Much like it sounds, scalar multiplication is responsible for *scaling* a vector to increase or decrease its magnitude, and the result is another vector. We now introduce an alternate version of multiplication, the dot product. Unlike scalar multiplication, **the dot product is applied to two vector quantities and produces a scalar quantity**. It is particularly useful when evaluating the angle formed between two vectors. We consider two definitions for the dot product, one from an algebraic perspective and the other from a geometric perspective.

DEFINITION: *The Dot Product (Algebraic Definition)*

Given two nonzero vectors, $\mathbf{v} = \langle a, b\rangle$ and $\mathbf{w} = \langle c, d\rangle$, the dot product of $\mathbf{v}$ and $\mathbf{w}$ is defined as follows.

$$\mathbf{v} \cdot \mathbf{w} = \langle a, b\rangle \cdot \langle c, d\rangle = ac + bd$$

EXAMPLE 6

(a) Let $\mathbf{v} = \langle 1, -2\rangle$ and $\mathbf{w} = \langle -2, 3\rangle$, evaluate $\mathbf{v} \cdot \mathbf{w}$.

$\langle 1,-2\rangle \cdot \langle -2,3\rangle = 1(-2) + -2(3) = -2-6 = -8$

(b) Evaluate $\hat{\imath} \cdot \hat{\jmath}$.

$\langle 1,0\rangle \cdot \langle 0,1\rangle = 1(0) + 0(1) = 0$

↑ important result!

Zero dot product → orthogonal vectors

DEFINITION: *The Dot Product (Geometric Definition)*

Given two nonzero vectors, $\mathbf{v} = \langle a, b\rangle$ and $\mathbf{w} = \langle c, d\rangle$, the dot product of $\mathbf{v}$ and $\mathbf{w}$ is defined as follows.

$$\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}||\mathbf{w}|\cos(\theta)$$

Here, θ is the angle formed between $\mathbf{v}$ and $\mathbf{w}$, with $0° \le \theta \le 180°$.

EXAMPLE 7

(a) Evaluate $\hat{\imath} \cdot \hat{\jmath}$.

$\langle 1,0\rangle \cdot \langle 0,1\rangle = |\langle 1,0\rangle|\,|\langle 0,1\rangle| \cos(90°) = (1)(1)(0) = 0$

← perpendicular (orthogonal)

↑ orthogonal vectors

(b) Let $\mathbf{v} = \langle 1, -2\rangle$ and $\mathbf{w} = \langle -2, 3\rangle$. What is the angle formed between these two vectors?

$\vec{v} \cdot \vec{w} = |\vec{v}||\vec{w}|\cos\theta \rightarrow \cos\theta = \dfrac{\vec{v} \cdot \vec{w}}{|\vec{v}|\,|\vec{w}|} = \dfrac{-8}{\sqrt{5}(\sqrt{13})}$

$|\vec{v}| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$

$|\vec{w}| = \sqrt{(-2)^2 + (3)^2} = \sqrt{13}$

$\theta = \cos^{-1}\left(\dfrac{-8}{\sqrt{65}}\right) \approx 172.87°$